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RV College of Engineering

(An Autonomous Institution Affiliated to VTU)

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III Semester B.E. Regular / Supplementary Examination February/March 2026

COMMON TO CD / CS / CY / IS

Operating Systems

Time : 3 Hours **Maximum Marks : 100**

Instructions to the students

1. Answer all questions from Part A . Part A questions should be answered in first three pages of the answer book only.
2. Answer Five full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8, 9 and 10.

Part A

Question No	Question	M	CO	BT
1.1	Classify the following as privileged/unprivileged. a) Modifying Interrupt Vector Table b) Reading system time using gettimeofday( );	02	1	2
1.2	Identify the multithreading model where a)The entire process blocks when a thread makes blocking system call b)The entire process does NOT blocks when a thread makes blocking system call	02	1	3
1.3	What do you mean by Spinlock?	02	3	2
1.4	A process of size 12 KB needs allocation. Blocks: 10 KB, 16 KB, 13 KB, 8 KB. Which block does First-fit, Best-fit and Worst-fit choose?	02	2	3
1.5	List any four fields stored in a File Control Block (FCB).	02	1	1
1.6	Why does a hybrid kernel offer better stability than a pure monolithic kernel?	02	1	2
1.7	Define Aging	02	2	2

1.8	Two processes P0 and P1 use two semaphores S and Q, both initialized to 1.
	a) P0 executes: wait(S); wait(Q);
	b) P1 executes: wait(Q); wait(S);
	If both processes acquire their first semaphore at the same time, what condition will the system enter?
		02	2	2

1.9	A system replaces pages aggressively but CPU utilization remains low. Identify the issue and suggest the solution to fix it.	02	1	2
1.10	Differentiate between superblock and inode.	02	1	1

Part B

Question No	Question	M	CO	BT
2a	A computer system is running three different processes: • Process A performs file reading. • Process B is executing a CPU-intensive calculation, and • Process C is waiting for network input. During execution, the operating system observes the following events: i) Process A issues a read request to the disk. ii)Process B's time quantum expires while it is still running. iii)The network packet that Process C was waiting for arrives. For each event, identify the process state transition that occurs and explain *why* the transition happens.	08	2	4
2b	Draw the queuing diagram of process scheduling and explain the different queues used by the operating system.	08	1	1
3a	Explain the difference between user threads and kernel threads.	06	1	2
3b	Consider the following scenario with processes and length of the CPU burst time given in milliseconds Process — Burst Time P1 — 3 P2 — 6 P3 — 4 P4 — 2 Draw Gantt charts illustrating the execution of these processes using FCFS and SJF. Compute the average waiting time, average turnaround time and number of context switches in each approach.	10	3	3
	OR			
4a	Analyze why real-time scheduling algorithms must guarantee deadlines. How do EDF and RMS achieve this?	06	3	4
4b	Write a C program using Pthreads where one thread reads integers from a file and stores them in an array, and another thread computes and prints the sum and average of the numbers. Include proper synchronization.	10	3	3
5a	A database allows multiple readers but only one writer at a time. Analyze this scenario and explain how semaphores can be used to implement the readers–writers problem.	08	4	4
5b	Write the pseudocode and the use of TestAndSet() and CompareAndSwap() hardware based synchronization primitives.	08	4	3
	OR			
6a	Illustrate the Test-and-Set operation used in hardware synchronization for solving the critical section problem. Explain how it ensures mutual	06	3	2

exclusion with a simple algorithm.

A software company is developing a real-time embedded system where two concurrent processes, P0 and P1, need to access a shared log file stored in memory. To prevent data inconsistency, the system designer decides to use Peterson's solution to handle mutual exclusion without relying on hardware support.

The following shared variables are used:

boolean flag[2] = {false, false};
int turn;

ach process follows Peterson's entry and exit protocol before accessing the critical section.

During testing, it is observed that:

6b

- Both processes repeatedly request access to the critical section. 10 3 4
- No process should starve or be indefinitely blocked.
- Only one process must be allowed in the critical section at any time.

Based on this scenario, answer the following:

i) Explain how Peterson's solution ensures mutual exclusion between P0 and P1.
ii) Describe the role of the variables `flag[]` and `turn` in resolving conflicts when both processes want to enter the critical section simultaneously.
iii) Justify how Peterson's solution satisfies the progress and bounded waiting requirements in this case.
iv) Mention one limitation of Peterson's solution when applied to modern multiprocessor systems.

7a	With the help of a neat diagram discuss the hardware support for implementation of page table. Consider a paging system with TLB. If it takes 20 ns to search TLB and 100 ns to access the memory what is the effective memory access time with 98-percent hit ratio?	10	3	3
7b	Calculate page faults using FIFO. The frame size is 3 and the Reference string is: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3	06	2	3
	OR			
8a	With the help of neat diagram discuss the logical address to physical address translation in paging memory management scheme. Mention it's merits and demerits in comparison with segmentation memory management scheme.	08	2	2
8b	Discuss the following concepts with respect to virtual memory in operating systems: Swap Space, Lazy Swapper, Pure Demand Paging, Copy-on-Write.	08	1	2
9a	Describe the different categories of files, their key attributes, and the operations supported on files.	06	1	1
9b	A multi-user operating system needs to restrict file access based on user permissions. Explain why NTFS is suitable and FAT is not.	10	3	2
	OR			
10a	Explain the different components of a typical File-System Layout.	06	1	2
10b	Compare FAT and NTFS filesystems.	10	1	2